Synthetic Respondents without Ambivalence?

A Population-Fidelity Audit of Large Language Model Survey Simulations

Abstract

Large language models can generate plausible survey answers without reproducing the population those answers are meant to represent. This article develops an estimand-indexed population-fidelity audit that separates marginal calibration, response heterogeneity, relational structure, and predictive benchmarking. A Monte Carlo study shows why the separation matters: generators with nearly identical univariate fit can produce radically different covariance structures. The audit is then applied to five repeated Qwen3-8B responses for each of 300 profiles sampled from three Chinese General Social Survey waves. The model shifts several item means, cuts response variance by more than half, and raises the housework item's mean absolute correlation with the other attitudes from .049–.078 among weighted human respondents to .437–.507. Across all three waves, every audited property falls outside a reference envelope generated by repeated human samples of the same size. Outcome-informed references remain imperfect for individuals, but a simple joint-donor model recovers both marginal and relational structure far better. Sparse profile information is therefore a real prediction limit, but it is not a sufficient explanation for the recorded language model's population distortion.

Keywords: synthetic respondents; large language models; survey simulation; population fidelity; machine learning; gender attitudes

1 Introduction

Large language models can produce answers that sound like survey responses from recognizable social types. Given a profile describing an older rural man with little education, a model can select a plausible position on a gender-role question; given a younger urban woman with a university education, it can generate a different one. This capacity has encouraged proposals to use language models as experimental subjects, pilot respondents, or even substitutes for expensive human samples (Aher et al. 2023; Argyle et al. 2023; Dillion et al. 2023). The methodological difficulty is that plausibility is an individual-level judgment, whereas survey inference concerns populations. Researchers need distributions, disagreement, covariance, subgroup contrasts, and uncertainty—not merely fluent impersonations.[1]

Existing audits establish that this distinction can fail in practice. Demographic conditioning can recover some subgroup patterns, the result described as "algorithmic fidelity" by Argyle et al. (2023). Yet language-model samples can also misrepresent opinion distributions, compress variance, change regression coefficients, react to small prompt changes, and drift when a proprietary model is updated (Bisbee et al. 2024; Qu & Wang 2024; Santurkar et al. 2023). These findings have shifted the question from whether an LLM can ever resemble a human sample to when resemblance is adequate for a stated analytical use.

Two gaps remain. First, evaluations often move directly from individual or marginal accuracy to a global verdict on "fidelity." That verdict obscures which property failed. A

[1]The term *synthetic respondent* here refers to an LLM prompted to answer as a persona. It should not be confused with conventional statistical synthetic data, which are generated from an explicitly estimated model of observed confidential data.

generator can match means while erasing covariance, or miss many individuals while still recovering a population distribution. Second, a failed persona prediction has at least two explanations. Demographic profiles may contain little information about a person's attitude, or the generator may distort the information that is available. Without an outcome-informed reference trained on the same profile features, it is difficult to determine whether sparse profiles are a sufficient explanation for the observed error.

This article critically assesses persona-based survey simulation and offers a usable correction: a four-stage *population-fidelity audit*. The audit evaluates (1) marginal calibration, (2) response heterogeneity, (3) relational structure, and (4) performance relative to transparent outcome-informed references. The stages are not a new estimator and do not establish a universal score. They are a decision procedure: researchers specify the downstream estimand, test every population property that estimand requires, and compare the synthetic error with a reference distribution obtained from equally sized human samples. A mean-estimation task may require marginal calibration; a regression, scale, or latent-class analysis additionally requires covariance and subgroup structure. The emphasis on a staged, reproducible workflow follows a broader computational-methods tradition in sociology (Nelson 2020).

The evidence has two parts. A Monte Carlo study calibrated to real survey marginals compares a row-bootstrap generator, an independent-marginals generator, and a deliberately coherent single-factor generator. All three achieve almost identical mean and distributional fit, but only the row bootstrap preserves the human correlation matrix. The simulation demonstrates the failure regime for mean-only validation rather than assuming it.

The real-data illustration uses 31,856 respondents from the 2012, 2018, and 2021 Chinese General Social Survey (CGSS) and 300 matched profiles sent to a locally deployed Qwen3-8B model. Five gender-attitude items create a demanding benchmark because human respondents combine public-sphere and household beliefs in ways that are not strongly unidimensional. The model is compared with the matched human answers, weighted population distributions, a marginal prior, multinomial logit, histogram gradient boosting, and a stratified joint-donor generator. These references receive no more profile features than the LLM at prediction time, although unlike the zero-shot LLM they learn from survey outcomes during training.[2]

The results are substantial and internally consistent for this case. Qwen3-8B's ordinal mean absolute error is 1.379 on the five-point scale, compared with 0.923 for histogram gradient boosting. Its average total variation distance is 0.538 rather than 0.071, and its variance ratio is 0.466 rather than 0.987. More consequentially, equal-housework attitudes have mean absolute correlations of .437–.507 with the other items, compared with .049–.078 in the weighted human data. All 15 wave-by-property audit results fall outside the 95 percent reference envelopes from repeated human samples. The model therefore creates not only a miscentered population, but a more homogeneous and ideologically coherent one.

The contribution is methodological and bounded. The article does not claim that all LLMs fail, that survey-trained prediction models explain human psychology, or that one Chinese gender battery represents all public opinion. It shows that validation must be indexed to population properties and that outcome-informed references can test whether limited profile information is a sufficient explanation for observed distortion. The design does not isolate model architecture from the absence of task-specific calibration. The accompanying code reproduces the simulation, figures, and empirical comparisons.

[2]The supervised comparison is intentionally asymmetric in training information. It is not a contest over equal training budgets. Its purpose is to determine whether sparse profile features impose the particular errors produced by the zero-shot generator.

2 Synthetic Respondents as a Measurement Procedure

2.1 The target is a distribution, not a persona

Let $\boldsymbol{X}_i$ denote the observed profile of respondent i, q_j a survey item, and $Y_{ij} \in \{1, \ldots, K\}$ the human response. A persona-prompted language model produces

$$\widetilde{Y}_{ij}^{L} = g_\theta(\boldsymbol{X}_i, q_j, p, s)\,, \tag{1}$$

where θ denotes the fixed model, p the prompt, and s the decoding seed. A fluent value of $\widetilde{Y}_{ij}^{L}$ can be plausible conditional on $\boldsymbol{X}_i$ without reproducing the human conditional distribution

$$\pi_{ijk} = \mathbb{P}(Y_{ij} = k \mid \boldsymbol{X}_i), \qquad k = 1, \ldots, K. \tag{2}$$

The gap between Equation (1) and Equation (2) is the central validation problem. Pretraining does not directly optimize g_θ for the target survey, population, language, year, response scale, or estimand. Prompting supplies context but not empirical base rates.

This formulation treats persona simulation as a measurement procedure. The generated response is an instrument reading produced by a model, prompt, and decoder. As with any instrument, validity is use-specific. Exact agreement with one respondent is demanding because survey answers contain idiosyncratic and transient variation. But low individual accuracy does not license arbitrary aggregate error. A model used to estimate a mean must recover the relevant marginal distribution; a model used for association or measurement analysis must also recover the joint distribution.

The distinction is increasingly important as LLMs enter social-science workflows (Chae & Davidson 2025; Demszky et al. 2023). Classification research can validate predictions against labels, but synthetic respondents are often invoked precisely where labels are scarce or absent. That makes explicit external benchmarks more important, not less. Otherwise, coherent output can create an illusion of understanding: the system produces a complete population because the researcher asked for one, not because that population has been measured (Messeri & Crockett 2024).

2.2 Four levels of population fidelity

Figure 1 summarizes the proposed audit. Each level corresponds to a different property of the population distribution and a different class of downstream estimands.

Marginal fidelity. The first level asks whether the generator recovers category shares, means, and other univariate features. These quantities are sufficient for a narrow descriptive estimand, but they say little about relationships among variables. For item j, total variation distance is

$$\mathrm{TV}_j = \frac{1}{2}\sum_{k=1}^{K}\left|\widehat{\pi}_{jk}^{L} - \widehat{\pi}_{jk}^{H}\right|, \tag{3}$$

where L and H denote language-model and human distributions.

Heterogeneity fidelity. The second level asks whether disagreement is preserved. The variance ratio

$$R_j^V = \frac{\mathrm{Var}(\widetilde{Y}_j^L)}{\mathrm{Var}(Y_j^H)} \tag{4}$$

equals one under variance recovery. Values below one indicate compression. A model may match a mean by placing nearly everyone near it, producing misleading standard errors, subgroup overlap, and latent-class prevalence.

Relational fidelity. The third level concerns the joint distribution. This article uses the root mean squared difference between item-correlation matrices,

$$\mathrm{RMSE}_R = \left[J^{-2} \sum_{j=1}^{J} \sum_{\ell=1}^{J} \left(\widehat{r}^{L}_{j\ell} - \widehat{r}^{H}_{j\ell} \right)^2 \right]^{1/2}, \tag{5}$$

and errors in demographic regression coefficients. Correlations are not privileged as causal or latent parameters; they are transparent diagnostics for whether synthetic data preserve relationships routinely used in social-science analysis.

Predictive benchmarking. The fourth level asks whether the generator adds value relative to explicit references. Let $p_{mij}(k)$ be the probability assigned by model m and $\widehat{\mu}_{mij} = \sum_k k p_{mij}(k)$. Ordinal mean absolute error is

$$\mathrm{OMAE}_m = \frac{1}{NJ} \sum_{i=1}^{N} \sum_{j=1}^{J} \left| \widehat{\mu}_{mij} - Y_{ij} \right|. \tag{6}$$

For the LLM, the primary analysis uses $R = 5$ repeated draws,

$$\widehat{\mu}_{Lij} = R^{-1} \sum_{r=1}^{R} \widetilde{Y}^{L}_{ijr}, \tag{7}$$

while a single-draw version remains a sensitivity analysis. References should include a marginal prior, an interpretable conditional model, a flexible learner, and—when relational fidelity is required—a generator that produces joint outcomes. This ladder distinguishes base-rate calibration from personalization, nonlinear prediction, and joint reconstruction.

The deployment rule is therefore conditional rather than universal: a synthetic sample must pass the stages required for its intended estimand and outperform a simpler reference that has comparable access to prediction-time information. In the empirical audit, “pass”

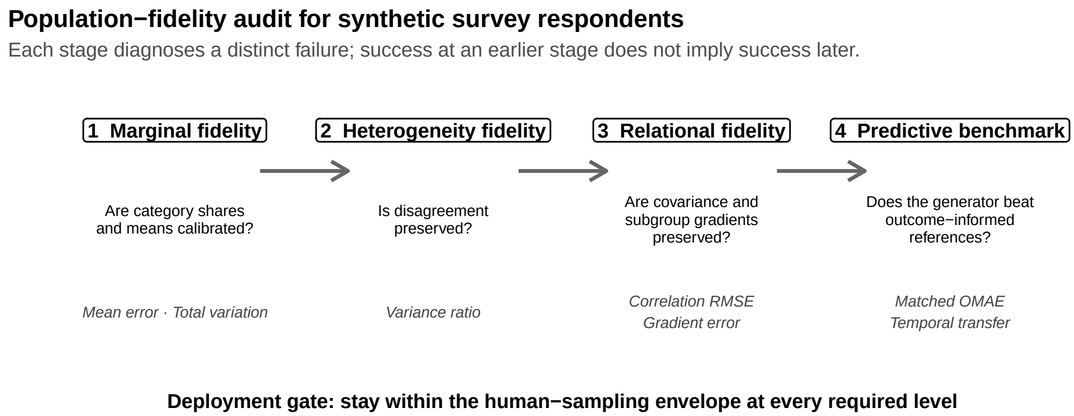

Figure 1: Population-fidelity audit for synthetic survey respondents. The stages are cumulative only relative to an intended estimand: passing marginal calibration does not establish heterogeneity or relational fidelity.

means that the synthetic error remains inside the central 95 percent reference envelope obtained by repeatedly drawing human samples with the same profile-sampling design. Failure at a required stage is not repaired by fluency at the individual level.

2.3 Why gender attitudes are a hard structural test

The empirical illustration uses gender attitudes because meaningful inconsistency is common. Gender operates across institutional, interactional, and individual settings (Risman 2004). Support for women's education or employment can coexist with traditional expectations inside marriage. Work on the uneven gender revolution and multiple egalitarianisms therefore treats ambivalence and domain differentiation as substantive patterns, not merely noise (England 2010; Knight & Brinton 2017; Scarborough et al. 2019).

The Chinese case sharpens this test. Public participation by women has long had institutional legitimacy, while domestic labor remains strongly organized through households. Research using Chinese surveys accordingly distinguishes work and family spheres and documents variation across cohorts, regions, gender, and education (Qian & Li 2020; Shu 2004;

Yang 2023). A language model may instead map a profile to a compressed social type—"traditional" or "modern"—and carry that type across every item. If so, the synthetic data will look more unidimensional than the human data.

This battery is not a definitive measurement model. Four items are traditional statements and one asks positively about equal housework. The housework item is a single indicator, so weak covariance can reflect content, wording, acquiescence, or item-specific error.[3] This limitation makes the test more conservative: the LLM is evaluated on whether it reproduces the empirical structure, not on whether one theoretical explanation for that structure is true.

3 Audit Design

3.1 Reference ladder

Table 1 makes the comparison logic explicit. The weighted prior observes outcome frequencies but ignores profiles. Multinomial logit estimates additive conditional associations. Histogram gradient boosting (HGB) permits nonlinearities and interactions (Mullainathan & Spiess 2017; Pedregosa et al. 2011). A stratified joint-donor generator samples complete five-item response vectors from training respondents matched on wave, sex, education group, and urban residence; it is deliberately simple but preserves respondent-level dependence. Qwen3-8B observes no task-specific survey outcomes and is evaluated zero-shot. The references are not psychological theories. They diagnose what can be reconstructed when survey outcomes are available.

[3] The audit therefore uses the observed weak association as a benchmark property to be reproduced. It does not claim that one item identifies a complete household-gender latent variable.

Table 1: Reference ladder for diagnosing synthetic-respondent failure

Method	Outcome information	Profile information	Diagnostic role	Main limitation
Weighted marginal prior	Training-fold category shares	None	Base-rate calibration without personalization	Cannot recover subgroup differences
Multinomial logit	Training-fold individual outcomes	Same features shown to the LLM	Transparent conditional benchmark	Additive linear predictor on the log-odds scale
Histogram gradient boosting	Training-fold individual outcomes	Same features shown to the LLM	Flexible outcome-informed benchmark	Task-specific and outcome-informed
Stratified joint donor	Complete training-fold response vectors	Wave, sex, education group, and residence	Transparent benchmark for joint reconstruction	Coarse matching and task-specific outcomes
Qwen3-8B	No CGSS outcomes	Persona text built from the same features	Zero-shot test of implicit base rates and conditioning	Model-, prompt-, and decoding-specific

Notes: "Same features" refers to survey wave, province, sex, age, education, hukou, urban residence, partnership status, and relative economic position. The comparison equalizes prediction-time covariates, not training information.

3.2 Monte Carlo study

The simulation demonstrates why the first two audit stages cannot establish the third. The target population is calibrated to the weighted marginal distributions and joint response patterns of complete cases in CGSS 2021. Each replication draws $n = 300$ observations under three data-generating procedures:

1. **Row bootstrap**: sample complete respondent rows with probability proportional to the survey weight. This preserves the empirical joint distribution in expectation.
2. **Independent marginals**: draw each item independently from its weighted empirical category probabilities. Marginal distributions are correct, but covariance is removed.
3. **Coherent factor**: draw five latent normal variables with a common-factor correlation of .65 and discretize them using the empirical marginal thresholds. Marginals remain correct while coherence is deliberately inflated.

The design uses 1,000 replications with a fixed seed. Performance is measured by mean absolute item error, average item-level total variation, mean variance ratio, correlation-matrix RMSE, and the mean absolute correlation between the housework item and the first four items.[4] The row bootstrap is an empirically calibrated positive control, not a proposed replacement for a survey. The boundary cases are the independent and coherent generators: both should pass marginal tests while failing for opposite reasons at the relational stage.

3.3 Uncertainty and evidentiary status

The simulation results are finite-sample regularities, not analytical theorems. Monte Carlo standard errors are reported in the online supplement. Empirical uncertainty comes from

[4]For item j, total variation is $\frac{1}{2}\sum_{k=1}^{5}|p_{jk}^{\text{synthetic}} - p_{jk}^{\text{human}}|$. It ranges from zero for identical category distributions to one for disjoint distributions.

2,000 profile-cluster bootstrap replications within survey wave. A sampled profile carries all five items and all repeated model draws with it; neither items nor repeats are resampled independently. Intervals therefore describe uncertainty conditional on the matched 300-profile design and the fixed full-sample human benchmark. They do not reproduce every stage of the original multistage survey sample.

A separate calibration exercise repeats the exact stratified, probability-proportional-to-weight profile draw 1,000 times within each wave, with $n = 100$ each time. Every sampled human population is compared with the full weighted benchmark using the five audit metrics. The central 95 percent range of those errors is a sampling reference envelope, not a confidence interval for the finite CGSS sample. It operationalizes the deployment rule by showing how far an equally sized human sample would ordinarily depart from the survey benchmark.

4 Empirical Illustration: Gender Attitudes in China

4.1 Human benchmark

The human benchmark uses CGSS 2012, 2018, and 2021 (National Survey Research Center at Renmin University of China 2021). CGSS is a repeated cross-sectional, multistage probability survey of adults in China. The analysis requires a valid housework response, at least three valid responses among the other four items, complete profile variables, and a positive survey weight. The resulting samples contain 11,668 respondents in 2012, 12,478 in 2018, and 7,710 in 2021, totaling 31,856.

The items ask whether respondents agree that (A421) men should focus on careers and women on family, (A422) men are naturally more capable than women, (A423) marrying well is better for a woman than doing well, (A424) women should be dismissed first during an economic downturn, and (A425) spouses should share housework equally. Responses range from 1 (completely disagree) to 5 (completely agree). A421–A424 are reverse-coded as $6 - Y$, while A425 remains unchanged, so every reported score runs from 1 to 5 with higher values indicating greater egalitarianism.

The full weighted sample establishes the population benchmark for means, category shares, variances, and Pearson correlations. Unweighted Pearson and polychoric correlations are reported as sensitivity checks. Survey weights are also used for human regression gradients because the target is the sampled population rather than an unweighted convenience sample (Solon et al. 2015). The analyses are descriptive; references to demographic "effects" are avoided because the repeated cross-sections do not identify causal effects.[5]

4.2 Matched persona sample

One hundred respondents are selected from each wave. Sampling is stratified by sex, education group, and urban residence. Stratum allocations follow weighted population shares, and respondents are selected within strata with probabilities proportional to their survey weights. Each profile contains survey year, province, sex, age, years of education, hukou, urban residence, partnership status, and relative household economic position. The model-facing file excludes all five human answers.[6]

The design therefore supports three comparisons: the synthetic sample against the full weighted population, synthetic answers against the same respondents' answers, and synthetic subgroup patterns against those estimated in the full sample. Matching does not turn the observed answer into a perfectly measured latent attitude. It makes the information set

[5] Sex, education, age, hukou, and residence are treated as conditioning variables and social locations, not randomized treatments.

[6] "Matched" means that the LLM, human benchmark, and out-of-fold reference predictions are evaluated on the same 300 respondent records. It does not refer to propensity-score matching or to pairing different respondents on observed covariates.

and evaluation target auditable.

4.3 LLM generation and prompt ablation

The audit uses the open-weight Qwen3-8B model (Qwen Team 2025), executed locally through Ollama 0.31.1 on an Apple M4 computer with 24 GB of unified memory. The exact local model digest is recorded in the supplement. The model runs in non-thinking mode with temperature .7, top-p .9, and a recorded seed. The main Chinese prompt asks the model to answer as the described respondent, gives verbal labels and their unique numerical mapping, and requires a JSON object. Responses are accepted only when all five keys are present, scores fall between 1 and 5, and verbal labels agree with numeric values.

The primary joint condition generates five independently seeded response vectors for every one of the 300 profiles. These draws define an empirical predictive category distribution for each profile. The first draw is retained as a prespecified sensitivity analysis so that revised estimates remain comparable with the original pilot. A second full condition uses the same profiles but tells the model not to choose politically correct or socially expected answers and omits the verbal-label validation format. Because those two features change together, that contrast remains a prompt ablation rather than a factorial causal test.

A targeted follow-up separates joint from independent item presentation. Each item is sent through a fresh API call with the complete persona repeated, no conversation history, and no access to the other questions or prior answers. Before the full follow-up run, the primary contrast was fixed as

$$\Delta_r = \overline{|r|}_{\text{joint}} - \overline{|r|}_{\text{independent}},$$

where each term averages the absolute correlations between A425 and A421–A424. The joint condition has five repeats, whereas the independent condition has one response per profile and item. The profile-bootstrap interval therefore captures profile composition but not repeated-decoding variation in the independent condition; the contrast is an exploratory presentation test rather than a complete mechanism experiment. An interval for Δ_r entirely above zero would support context-induced coherence. If independent presentation remains above the CGSS benchmark, the two mechanisms may coexist.

4.4 Survey-trained references

The prior, multinomial logit, and HGB models use the same nine profile features. Numeric predictors are median-imputed and standardized; province and other categorical predictors are one-hot encoded. Training weights are normalized to mean one. Hyperparameters are fixed before matched-profile evaluation.

The main comparison uses five-fold stratified out-of-fold prediction for each item. Each person's outcome is excluded from the model that predicts that person. The 300 LLM profiles are joined to these predictions by exact wave and row identifiers. A second design trains on 2012 and 2018 and tests on 2021. This temporal transfer deliberately removes access to the target year's outcomes, although the reference models still observe historical labels.

The stratified joint-donor benchmark excludes the matched respondent, locates complete training respondents in the same wave-by-sex-by-education-by-residence cell, and samples five complete response vectors using survey weights. Its purpose is not optimal prediction. It tests whether a transparent outcome-informed procedure can preserve dependence that separate item classifiers cannot generate.

For every generator, the primary population moments are calculated from its predictive category distribution. The reference models supply probabilities directly; Qwen3-8B supplies a five-draw empirical distribution. Total Qwen variance is decomposed into variance between profile-specific means and average stochastic variance within profiles. The original one-draw variance ratio remains available as a sensitivity result rather than serving as the primary comparison.

4.5 Relational diagnostics

Item structure is summarized by correlation-matrix RMSE and by the mean absolute correlation between A425 and A421–A424. Human targets use survey-weighted Pearson correlations; unweighted Pearson and polychoric estimates test whether the weak A425 association is a weighting or ordinal-scaling artifact. Each of the five joint LLM draws produces a separate 100-profile correlation matrix, and the resulting diagnostics are averaged rather than treating 500 repeated rows as independent. Subgroup fidelity is assessed with parallel linear regressions for a four-item public-attitude index and A425. Each outcome is regressed on sex, years of education, age in decades, and urban residence. Human regressions use survey weights; synthetic regressions use the sampled profiles with equal weight because weighting already enters profile selection. Coefficient RMSE and sign agreement summarize recovery. These regressions are diagnostics of conditional structure, not substantive causal models.

5 Results

5.1 Mean-only validation accepts structurally wrong populations

Panel A of Table 2 reports the Monte Carlo study. All three generators have mean absolute item error between .0546 and .0559, average total variation between .0424 and .0435, and a mean variance ratio near one. On these metrics, the independent and coherent generators are indistinguishable from the row bootstrap.

The relational audit reverses that conclusion. Correlation-matrix RMSE is .0505 under row bootstrapping, .274 under independent marginals, and .371 under the coherent-factor generator. The housework item's mean absolute correlation is .0905 under the row bootstrap, .0457 under independent sampling, and .525 under the coherent factor. Mean-only validation would accept all three populations despite opposite and substantively consequential errors in their joint structure.

Table 2: Core simulation and matched-profile results

Generator or model	OMAE / mean error	Total variation	Variance ratio	Correlation RMSE	A425 mean $\lvert r \rvert$
Panel A. CGSS-calibrated Monte Carlo, 1,000 replications					
Row bootstrap	.0547	.0426	.997	.0505	.0905
Independent marginals	.0546	.0424	1.000	.274	.0457
Coherent factor	.0559	.0435	.999	.371	.525
Panel B. Same 300 empirical profiles					
Weighted marginal prior	.975	.0965	.992	—	—
Multinomial logit	.935	.0814	.982	—	—
Histogram gradient boosting	.923	.0712	.987	—	—
Stratified joint donor	.987	.0699	.973	.0851	.105
Qwen3-8B, neutral prompt	1.379	.538	.466	.293	.468

Notes: Panel A's second column is mean absolute item-mean error; Panel B's is individual ordinal mean absolute error. Panel A metrics are replication means. Correlation entries average the three wave-specific values. The prior, logit, and gradient-boosting models generate separate item probabilities, so relational metrics are undefined for them; the donor and Qwen procedures generate complete response vectors.

Panel D of Figure 3 visualizes the failure. The three scenarios occupy nearly the same position on total variation, while their relational error differs by a factor of more than six. This is the simulation-backed justification for treating relational fidelity as a separate audit stage.

5.2 Observed errors exceed ordinary human sampling variation

Table 3 turns the audit from a metric list into a calibrated decision rule. Across 1,000 repeated human samples per wave, ordinary samples of 100 respondents have average total variation below .105 at the upper edge of the central 95 percent reference envelope and correlation-matrix RMSE below .142. Their variance ratios remain between .846 and 1.141.

Qwen3-8B is outside the corresponding envelope for every metric in every wave—15 of 15 comparisons.

Table 3: Qwen3-8B relative to equally sized human sampling envelopes

Audit metric	Human 95% reference range	Qwen range	Waves outside
Absolute item-mean error	[.036, .168]	[.880, 1.021]	3 / 3
Total variation	[.046, .105]	[.514, .571]	3 / 3
Variance ratio	[.846, 1.141]	[.441, .592]	3 / 3
Correlation-matrix RMSE	[.047, .142]	[.279, .298]	3 / 3
A425 mean absolute correlation	[.034, .235]	[.437, .507]	3 / 3

Notes: Each wave contributes 1,000 stratified, probability-proportional-to-weight human samples of $n = 100$. The displayed human range runs from the smallest wave-specific 2.5th percentile to the largest wave-specific 97.5th percentile, making it a conservative union across waves rather than a confidence interval. Qwen uses five repeated draws per sampled profile.

The comparison does not imply that CGSS is error-free. It asks a narrower question: whether Qwen's departure could plausibly be attributed to using only 100 profiles per wave. The answer is no for all five audited properties.

5.3 Qwen3-8B misses item-specific opinion

Figure 2 shows that error is neither small nor a uniform ideological shift. Under the neutral prompt, Qwen3-8B approximates the family-role item and produces more egalitarian responses on male ability. It produces substantially more traditional responses on marriage, employment protection, and equal housework. Averaged across waves against the full weighted CGSS distributions, absolute item-mean error is .942 and total variation is .542.

The largest errors recur across waves. Support for equal housework is understated by 1.61 points in 2012, 1.43 in 2018, and 1.94 in 2021. Opposition to dismissing women first is understated by 1.47, 1.31, and 1.49 points. These differences generate a synthetic population that is much less supportive of household equality and employment protection

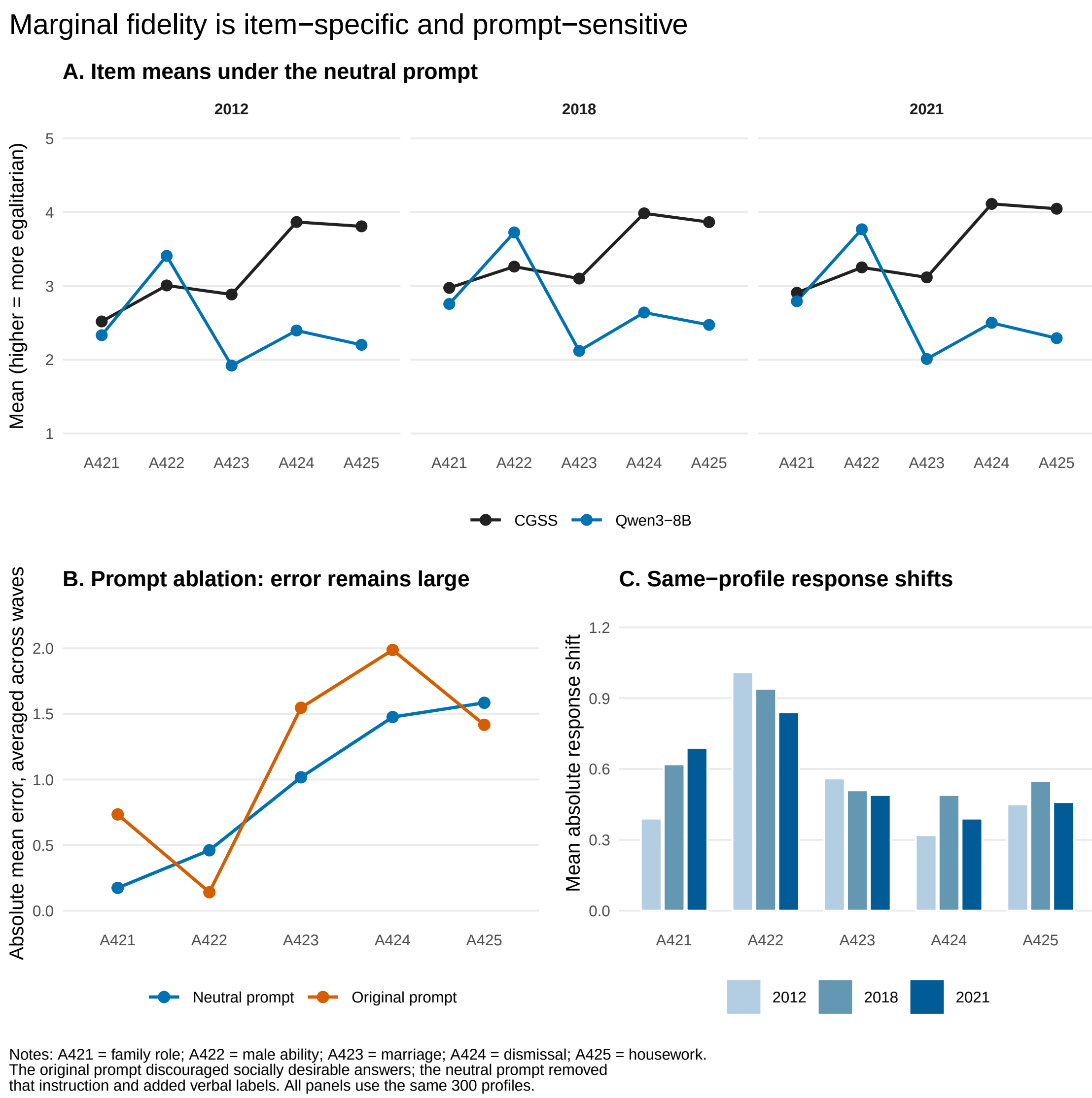

Figure 2: Marginal fidelity and prompt ablation. All scores are oriented so that higher values indicate greater gender egalitarianism. The original and neutral conditions contain the same 300 profiles; the ablation changes both the social-desirability instruction and response format.

than the corresponding survey populations.

Removing the anti-social-desirability instruction improves but does not repair the distribution. Mean absolute item error falls from 1.17 under the original prompt to .942 under the neutral prompt; average total variation falls from .664 to .542. Yet the same-profile mean absolute response shift is .581 points, demonstrating substantial prompt dependence. Prompt choice changes the severity and item pattern of error without yielding a condition that passes marginal calibration.

5.4 The synthetic population is too homogeneous and too coherent

Panels A and B of Figure 3 show two distinct structural failures. Across the 15 wave-item cells, the median Qwen-to-human total variance ratio is .412. Within-profile stochastic variation supplies about 23 percent of Qwen's predictive variance on average, so repeated sampling adds dispersion without closing the human gap. Compression is most pronounced for marriage and equal housework in 2012 and 2021. The model is not simply drawing a different population center; it is suppressing disagreement around that center.

Repeat stability is high but not deterministic. Across waves, 94 percent of A421 profiles and 80 percent of A422 profiles give the same answer in all five draws; the corresponding shares are 69 percent for A423, 58 percent for A424, and 72 percent for A425. The variance decomposition therefore captures genuine sampling variation, but most of the missing population heterogeneity lies between profiles rather than within repeated calls.

The model also imposes a joint ideology that the human data do not contain. In the weighted CGSS benchmark, the mean absolute correlation between A425 and A421–A424 is .049 in 2012 and .078 in both 2018 and 2021. The corresponding Qwen3-8B values are .460, .507, and .437, absolute gaps of .359–.429. Unweighted Pearson estimates are .051–.067, and unweighted polychoric estimates are .069–.091, so neither weighting nor ordinal

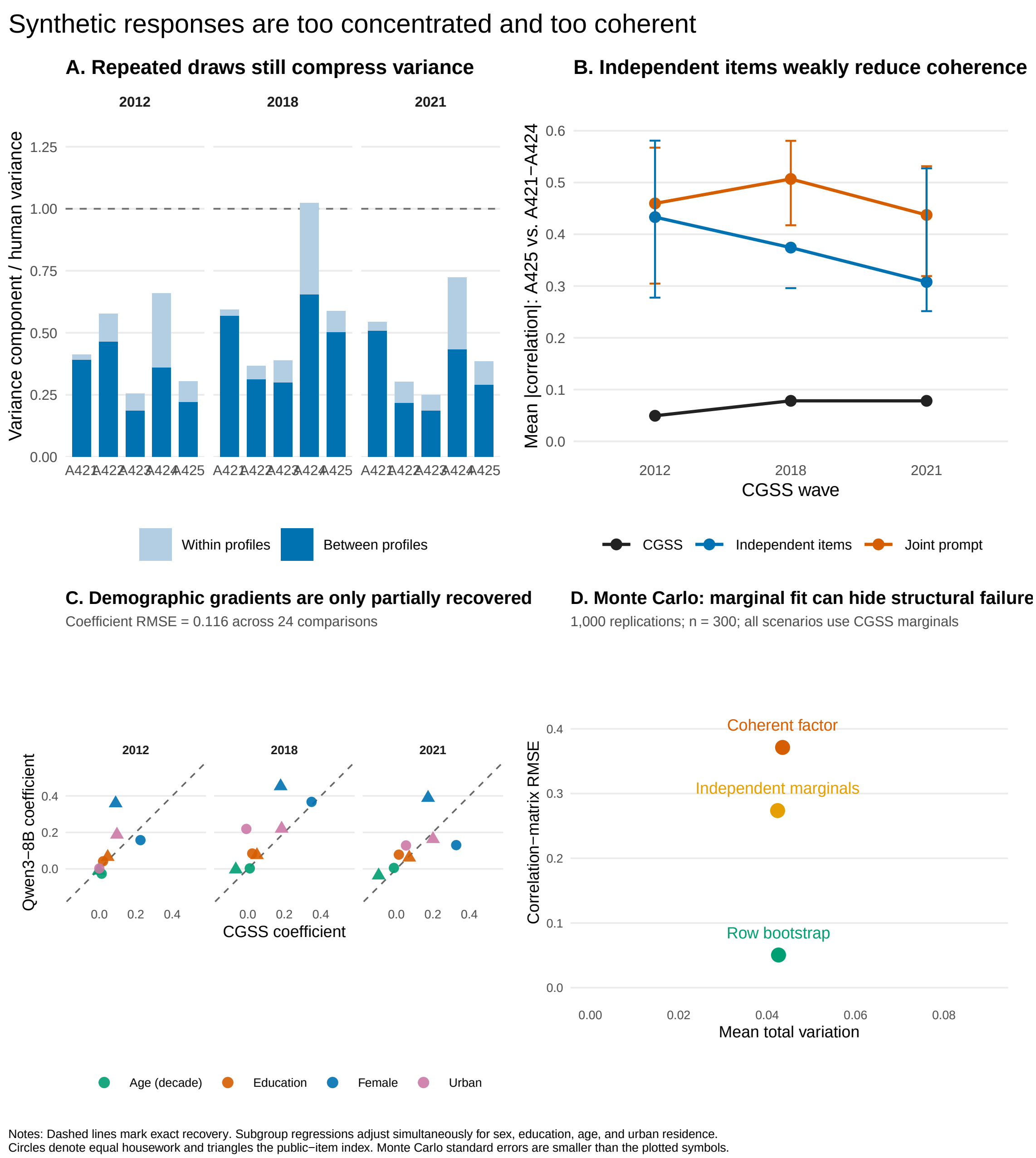

Figure 3: Heterogeneity and relational fidelity. Panel C compares descriptive demographic coefficients for the public-item index and equal housework. Panel D reports the CGSS-calibrated Monte Carlo scenarios.

scaling explains the synthetic coherence.

Demographic gradients are partly, not fully, recovered. Across 24 coefficient comparisons for sex, education, age, and urban residence, sign agreement is 83.3 percent and coefficient RMSE is .117. Education gradients are often directionally similar, but the model overstates several sex differences and misses some item-specific urban and household patterns. A model can therefore produce broadly recognizable demographic sorting while still misrepresenting its magnitude and domain specificity.

Independent presentation yields a smaller point estimate of excess coherence, but not a decisive mechanism test. The mean joint-minus-independent contrast is .096, with a profile-bootstrap 95 percent interval of $[-.043, .156]$. Independent-item mean absolute correlations remain .433, .374, and .308 across the three waves—still far above the weighted human values—while A422 and A424 become constant or nearly constant in several wave-item cells. The direction is compatible with context-induced consistency, but the interval includes zero, the independent condition has only one draw per profile-item, and the manipulation introduces a new heterogeneity failure. The defensible conclusion is therefore narrower: the recorded presentation procedures differ, but the experiment does not identify a unique internal mechanism and neither presentation recovers the observed population structure.

5.5 Sparse profiles do not explain the full error

Panel B of Table 2 and Panel A of Figure 4 address the main alternative explanation. The supervised models remain imperfect at the individual level: HGB's OMAE is .923 and multinomial logit's is .935. Demographic profiles therefore impose a genuine information ceiling. But the reference models recover aggregate distributions far better than Qwen3-8B. HGB's total variation is .071 and variance ratio .987, compared with .538 and .466 for

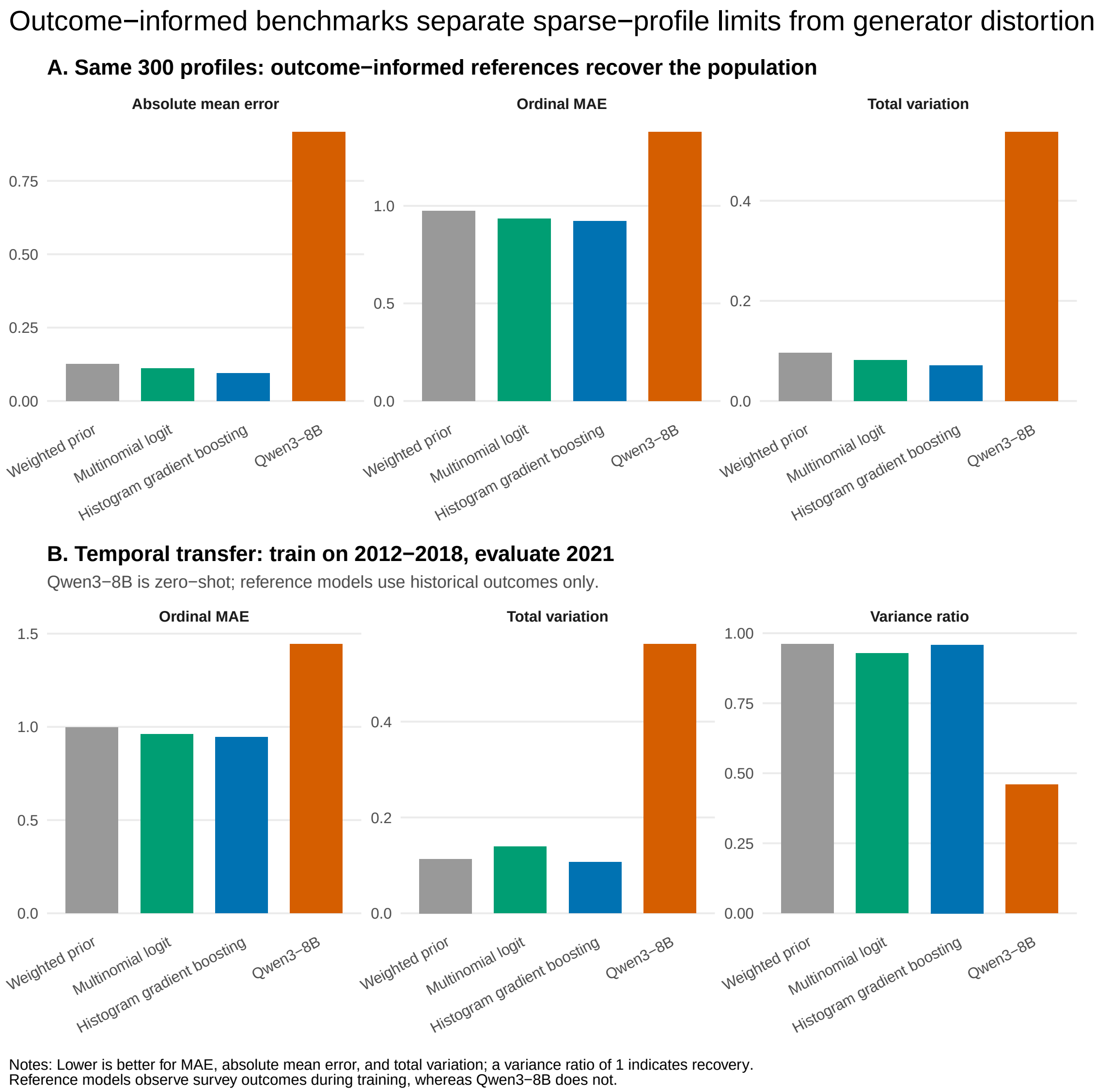

Figure 4: Matched-profile and temporal-transfer benchmarks. Lower values are better for ordinal MAE, absolute mean error, and total variation; one is the target for the variance ratio. The references are outcome-informed, whereas Qwen3-8B is zero-shot.

Qwen3-8B.

Even the weighted prior, which learns no mapping from profiles to answers, outperforms the LLM: OMAE is .975, total variation .097, and variance ratio .992. This does not make the prior information-free; it uses training-fold outcome frequencies. Its diagnostic value is that accurate base rates alone recover more of this population than demographic conditioning through the zero-shot LLM.

The joint-donor benchmark addresses the relational gap left by separate item classifiers. Despite matching only on wave, sex, education group, and residence, it has OMAE .987, total variation .070, variance ratio .973, correlation-matrix RMSE .085, and A425 mean absolute correlation .105. It is not an optimal model, but it shows that preserving complete response vectors substantially restores the human joint structure. The contrast is therefore not between an unattainable oracle and the LLM.

The paired individual comparison is also stable within the matched sample. Qwen3-8B's five-draw profile-mean absolute ordinal error exceeds HGB's by .456 points, with a profile-cluster bootstrap interval of $[.407, .507]$. The gap is concentrated in employment protection and equal housework rather than spread uniformly over the battery.

Temporal transfer provides a harder reference. On the exact 100-profile 2021 cohort used for the LLM audit, training on 2012 and 2018 gives HGB an OMAE of .945, total variation of .107, and variance ratio of .959. Public opinion shifts, and survey-trained models pay a penalty for extrapolation. Qwen3-8B on the same cohort has OMAE 1.44, total variation .561, and variance ratio .459. Historical outcome information is not equivalent to zero-shot generation, but the comparison shows that target-year exclusion alone does not produce the LLM's degree of distortion.

6 Discussion

6.1 What the audit identifies—and what it does not

The audit does not identify a pure architecture-level "generator effect." The outcome-informed references receive task-specific labels that the zero-shot LLM does not. What the audit establishes is narrower: sparse demographic profiles are not a sufficient explanation for the recorded distortion. Every model remains uncertain about individual gender attitudes, yet simple outcome-informed references recover the marginal distribution, and a stratified joint donor recovers much of the dependence structure. Sparse profiles therefore do not force a model to shift item means, halve population variance, or transform a weakly related housework item into part of a coherent ideology.

This distinction changes how synthetic respondents should be evaluated. Individual exact match, aggregate calibration, heterogeneity, and relational structure are different targets. No single accuracy score can certify them. The proposed procedure therefore begins with the intended estimand:

- For a population mean or category share, require marginal calibration and report uncertainty.
- For dispersion, polarization, or subgroup overlap, also require variance and tail recovery.
- For regressions, scales, latent classes, mediation, or networked attitudes, require joint and subgroup structure in addition to marginals.
- In every case, compare the LLM with a marginal prior and transparent task-specific learners so that failure is not attributed automatically to insufficient profile information.

- Calibrate tolerances against equally sized human samples rather than declaring success from an arbitrary numerical threshold.

The procedure is intentionally conservative. Passing does not prove that an LLM has learned the human mechanism; it shows only that the generated data reproduce stated benchmark properties within sampling-calibrated tolerances. Failing a required property is more informative: it rules out the generated sample for estimands that depend on that property.

6.2 Why synthetic coherence is substantively hazardous

In the gender-attitude illustration, inconsistency is part of the phenomenon. Human respondents may support public equality while assigning household obligations differently. The model replaces some of this ambivalence with a single demographic narrative. The result is a sample that appears easier to explain than the population it represents.

This error is particularly hazardous for computational social science. A researcher using the synthetic data could conclude that the five items form a stronger scale, identify cleaner ideological types, estimate steeper demographic gradients, or simulate more uniform attitude change. Each conclusion would be supported by orderly synthetic patterns that are weak or absent in the survey. The problem is not merely bias toward traditional or egalitarian answers. It is the production of a different social structure.

The stakeholders extend beyond the analyst. Policy organizations that substitute synthetic respondents for hard-to-reach populations may understate disagreement, erase minority combinations of beliefs, or design interventions around a consensus that does not exist. The harm is estimand-specific: mean distortion misstates prevalence, variance compression obscures pluralism, and synthetic coherence can manufacture a scale or social type. Choosing CGSS as the benchmark is also a normative and empirical decision. It privileges a probability survey with explicit weights over pretrained textual regularities, while acknowledging that the survey itself contains nonresponse, item error, and wording effects.

The finding also qualifies cross-cultural claims about LLM public-opinion simulation. Prior work documents uneven performance across countries and topics (Qu & Wang 2024). The present case adds a Chinese-language, open-weight model audit and shows why average country-level agreement is insufficient. Cultural or linguistic representativeness must be evaluated in the relationships among answers, not inferred from a plausible mean.

6.3 Comparison with prior synthetic-respondent audits

The design extends rather than contradicts existing evaluations. Argyle et al. (2023) show that demographic conditioning can recover nuanced response patterns in some settings. Bisbee et al. (2024) demonstrate failures in means, variances, conditional relationships, prompt sensitivity, and reproducibility. The present contribution adds two pieces. First, the Monte Carlo study makes explicit that marginal validation is non-identifying for joint structure. Second, the human-sampling envelope and outcome-informed reference ladder turn that insight into a falsification procedure. The answer in this application is partial: sparse persona attributes explain why no method predicts individuals perfectly, but they do not account for the recorded Qwen3-8B population. The design still cannot determine whether the remaining gap comes from model architecture, pretraining, prompt interpretation, or the absence of target-specific calibration.

This logic can travel beyond gender attitudes. Researchers evaluating synthetic respondents for health beliefs, consumer preferences, political ideology, or experimental behavior can replace the items and reference models while retaining the four-stage audit. The required relational diagnostic should follow the intended analysis: correlations for scale construction, treatment-effect heterogeneity for experiments, transition matrices for panels, or network statistics for social interaction.

7 Limitations and Required Next Tests

Seven limitations bound the evidence.

First, the empirical audit concerns one open-weight model and one parameter scale. Qwen3-8B is useful for reproducibility and local data protection, but a single model cannot support a claim about all LLMs. Cross-family and within-family scaling comparisons are the highest-priority extension.

Second, five draws provide an empirical predictive distribution, but they remain a coarse approximation to the model's option probabilities. They identify within-profile stochastic variation more credibly than a single draw, yet cannot support fine-grained entropy or calibration analysis. Direct candidate-token probabilities would provide a stronger probabilistic comparison.

Third, independent presentation changes the amount of questionnaire context by design and was generated only once per profile-item, whereas the joint condition has five repeats. The resulting contrast identifies neither a unique internal mechanism nor the full decoding uncertainty under independent presentation. Moreover, an item with nearly constant independent responses has no well-defined Pearson correlation; reduced correlation cannot then be interpreted as repaired relational fidelity without also checking heterogeneity.

Fourth, the original prompt ablation still changes both social-desirability wording and response format. It demonstrates sensitivity but does not identify which component causes it. A future factorial design should vary instruction, verbal labels, presentation mode, temperature, and seed separately.

Fifth, the survey battery has a measurement limitation. A425 is one positively worded item, whereas A421–A424 are traditional statements that are reverse-coded. Human weak covariance may reflect domain separation, wording, or both. Replication with multi-item public and household measures is needed before treating the observed structure as a general latent model of gender ideology.

Sixth, the sampling envelope treats the full weighted CGSS sample as the benchmark and reproduces the profile-selection stage rather than the original survey's complete multi-stage design. It calibrates finite-profile error but does not propagate all PSU, stratification, nonresponse, and weight-estimation uncertainty.

Seventh, the contents of Qwen3-8B's pretraining corpus are not fully observable. The model may have encountered the item wording, discussions of CGSS, or related normative language. Prompt paraphrases and temporal comparisons diagnose sensitivity, but they cannot identify contamination or memorization.

These limitations affect the breadth and mechanism of the claim, not the observed failure of the recorded Qwen3-8B procedure. The synthetic sample produced here does not recover the population properties required for scale, regression, or latent-structure analysis.

8 Conclusion

Synthetic respondents should be audited as populations. A plausible answer from a plausible persona does not establish correct base rates, disagreement, covariance, or subgroup structure. The population-fidelity audit turns that principle into a staged diagnostic tied to the intended estimand.

In a CGSS-calibrated simulation, generators with nearly indistinguishable marginal fit produce radically different joint distributions. In the empirical illustration, Qwen3-8B shifts several gender-attitude means, compresses variance, and imposes much stronger ideological coherence than appears among human respondents. Its errors lie outside the reference envelopes produced by equally sized human samples in every audited wave-property cell. Outcome-informed references confirm that demographic profiles contain limited individual information, but they also show that this limitation alone does not account for the LLM's specific population distortion.

The practical recommendation is direct: do not validate synthetic respondents with means alone, do not interpret low individual accuracy without an outcome-informed predictive benchmark, and do not use synthetic data for relational estimands until the relevant joint structure has been recovered.

Preregistration Statement

The study was not registered in a public repository before analysis. After the initial audit, but before completing the follow-up model calls, the repeated-draw estimand, profile-level bootstrap, independent-item manipulation, and sign of the joint-minus-independent correlation contrast were fixed in the dated analysis log. The follow-up is therefore specification-before-completion rather than a formal preregistration.

Code, Materials, and Data Availability

The code-only replication archive accompanying this writing sample contains the prompt templates, model-call runner, survey-trained benchmark code, Monte Carlo code, and aggregate results supporting the displayed analyses. Respondent-level CGSS data, derived persona files, matched human responses, and profile-level model logs are not part of the public release because access is governed by the data provider and the derived records remain linked to sampled profiles. Qualified researchers can obtain the source waves from the National Survey Research Center at Renmin University of China, rebuild the derived files locally, and regenerate the local-model calls. Before journal submission, the code-only review archive should be deposited in a blinded repository and assigned a permanent DOI upon acceptance.

Ethics and Consent Statement

The study is a secondary analysis of deidentified survey data and generated model outputs. No new human participants were recruited or contacted.

Declaration of Conflicting Interests

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Declaration of Generative AI Use

During preparation of this manuscript, the author used OpenAI Codex to assist with code review, figure production, manuscript restructuring, and language editing. The author reviewed the analyses, verified reported results against local outputs, edited the generated text, and takes full responsibility for the content.

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